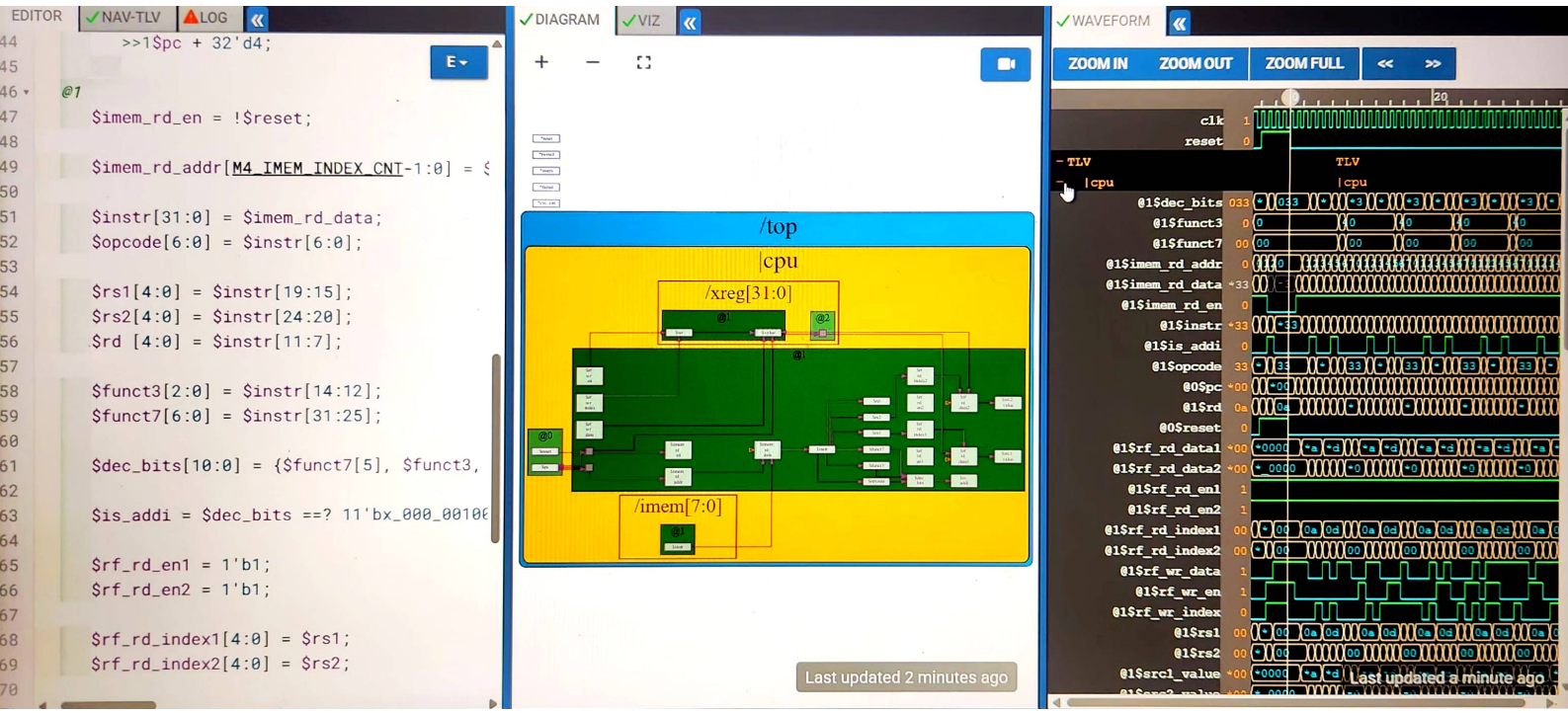
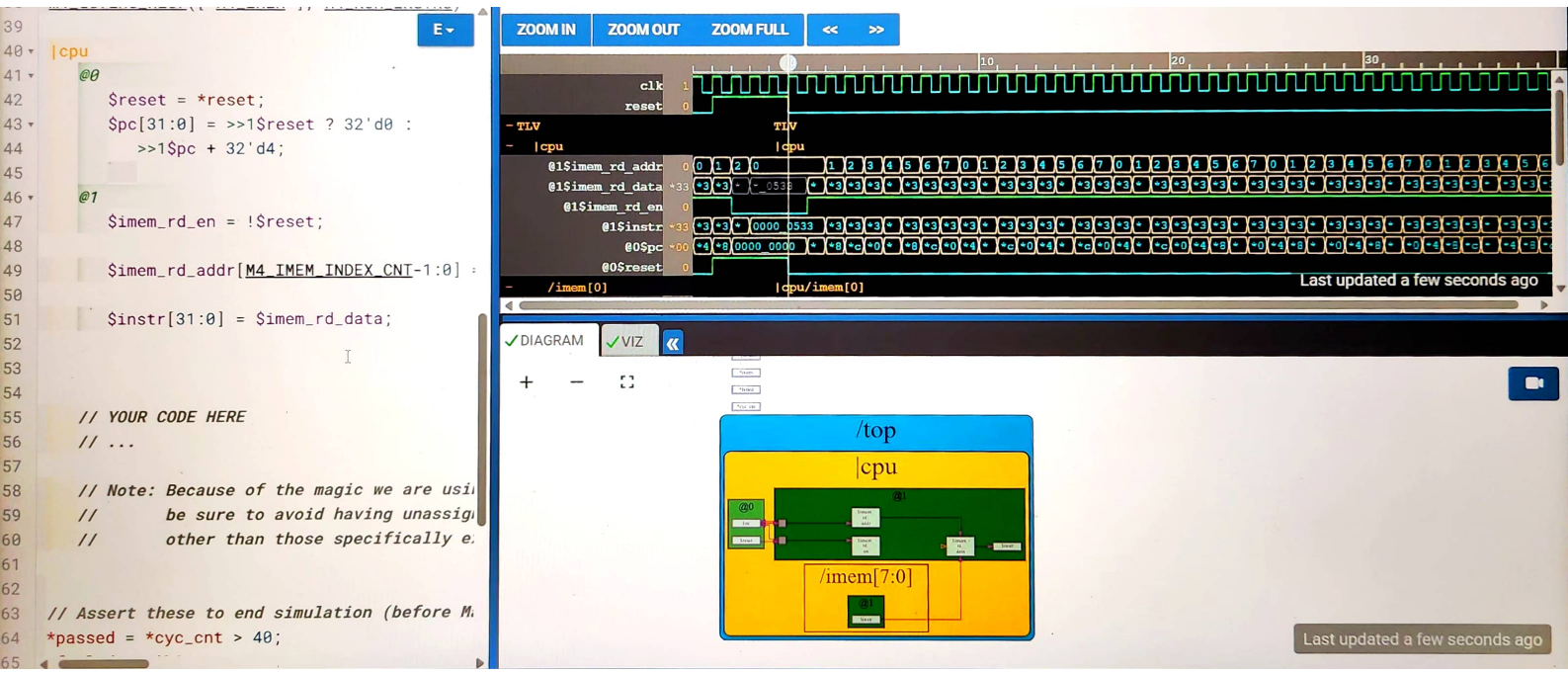
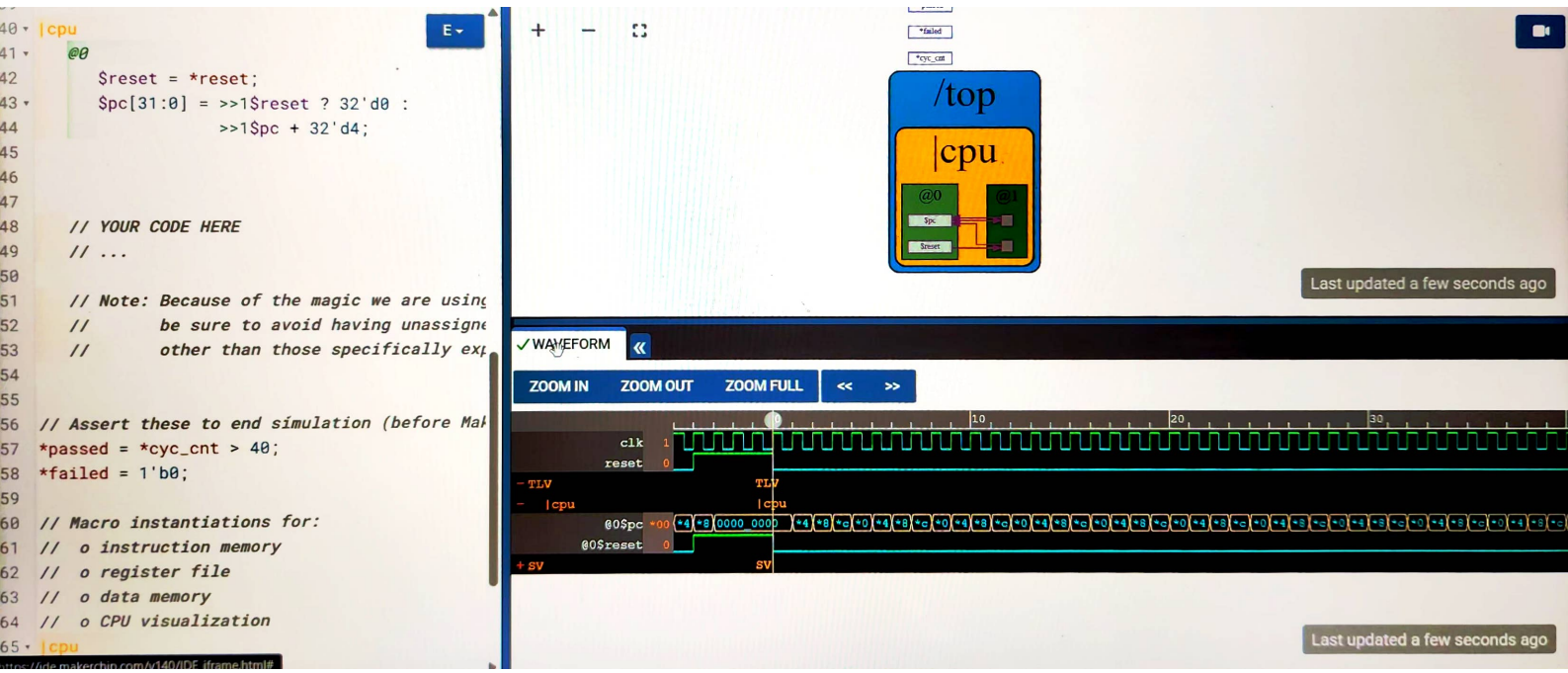
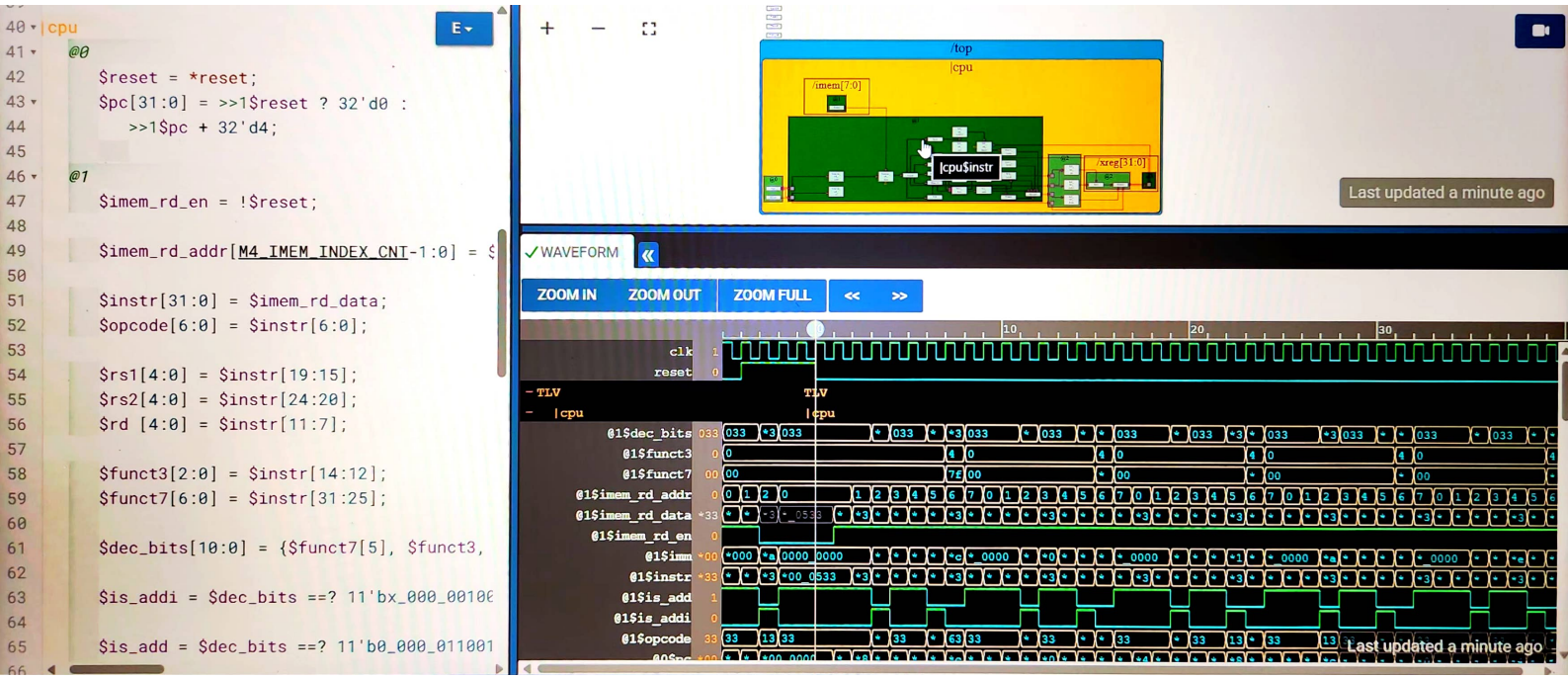
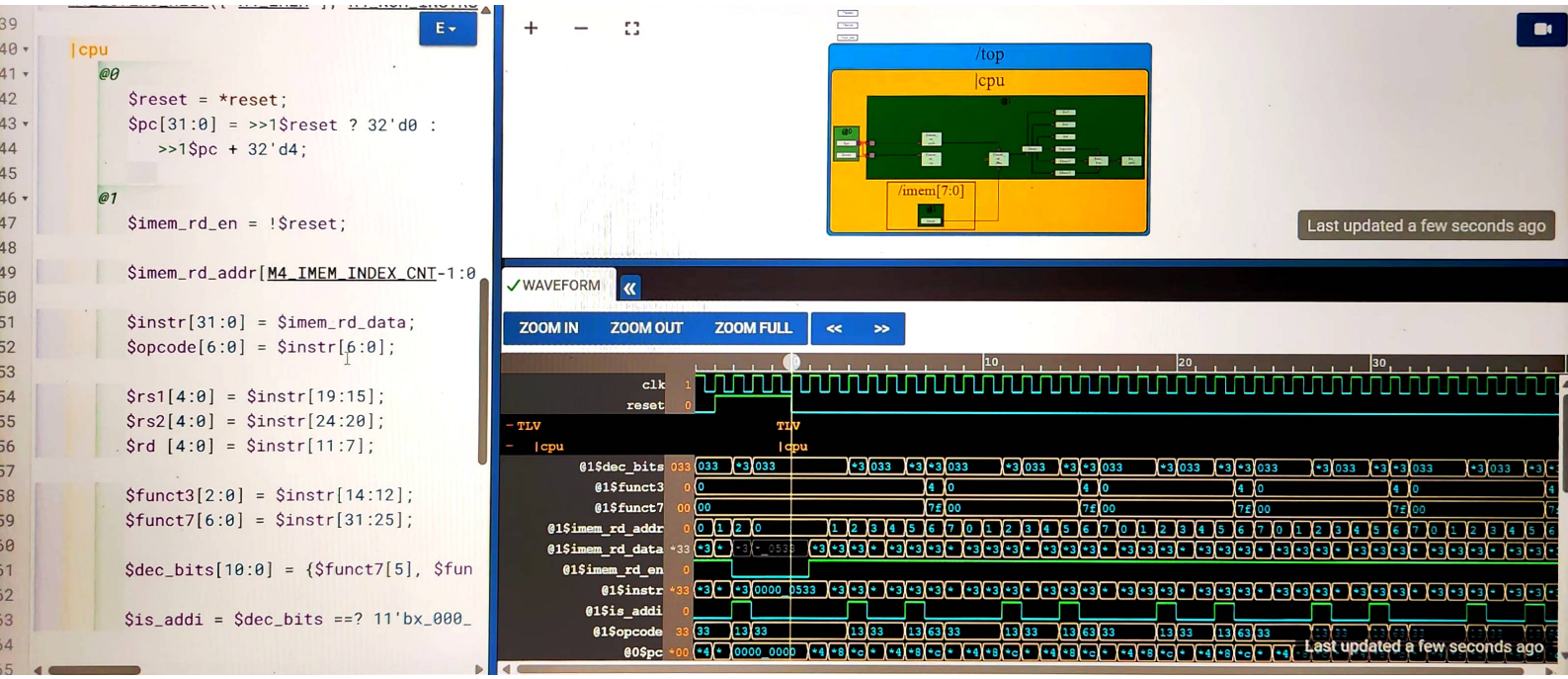
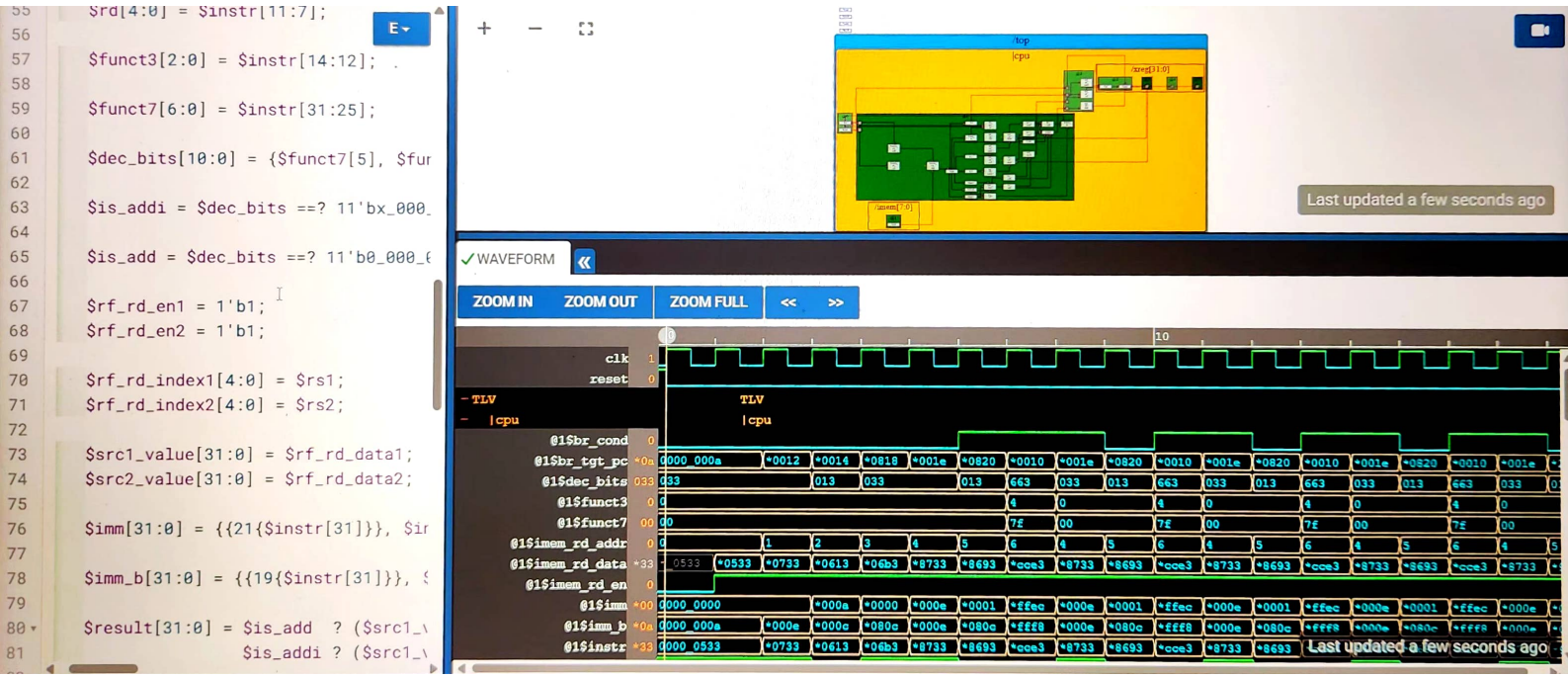

EDITOR
NAV-TLV
LOG
>>1$pc + 32'd4;
@1
$imem_rd_en = !$reset;
$imem_rd_addr[M4_IMEM_INDEX_CNT-1:0] = $
$instr[31:0] = $imem_rd_data;
$opcode[6:0] = $instr[6:0];
$rs1[4:0] = $instr[19:15];
$rs2[4:0] = $instr[24:20];
$rd [4:0] = $instr[11:7];
$funct3[2:0] = $instr[14:12];
$funct7[6:0] = $instr[31:25];
$dec_bits[10:0] = {$funct7[5], $funct3,
$is_addi = $dec_bits ==? 11'bx_000_00100
$rf_rd_en1 = 1'b1;
$rf_rd_en2 = 1'b1;
$rf_rd_index1[4:0] = $rs1;
$rf_rd_index2[4:0] = $rs2;
DIAGRAM
VIZ
/top
|cpu
/xreg[31:0]
/imem[7:0]
Last updated 2 minutes ago
WAVEFORM
ZOOM IN
ZOOM OUT
ZOOM FULL
clk
reset
TLV
|cpu
@1$dec_bits
@1$funct3
@1$funct7
@1$imem_rd_addr
@1$imem_rd_data
@1$imem_rd_en
@1$instr
@1$is_addi
@1$opcode
@0$pc
@1$rd
@0$reset
@1$rf_rd_data1
@1$rf_rd_data2
@1$rf_rd_en1
@1$rf_rd_en2
@1$rf_rd_index1
@1$rf_rd_index2
@1$rf_wr_data
@1$rf_wr_en
@1$rf_wr_index
@1$rs1
@1$rs2
@1$src1_value
Last updated a minute ago

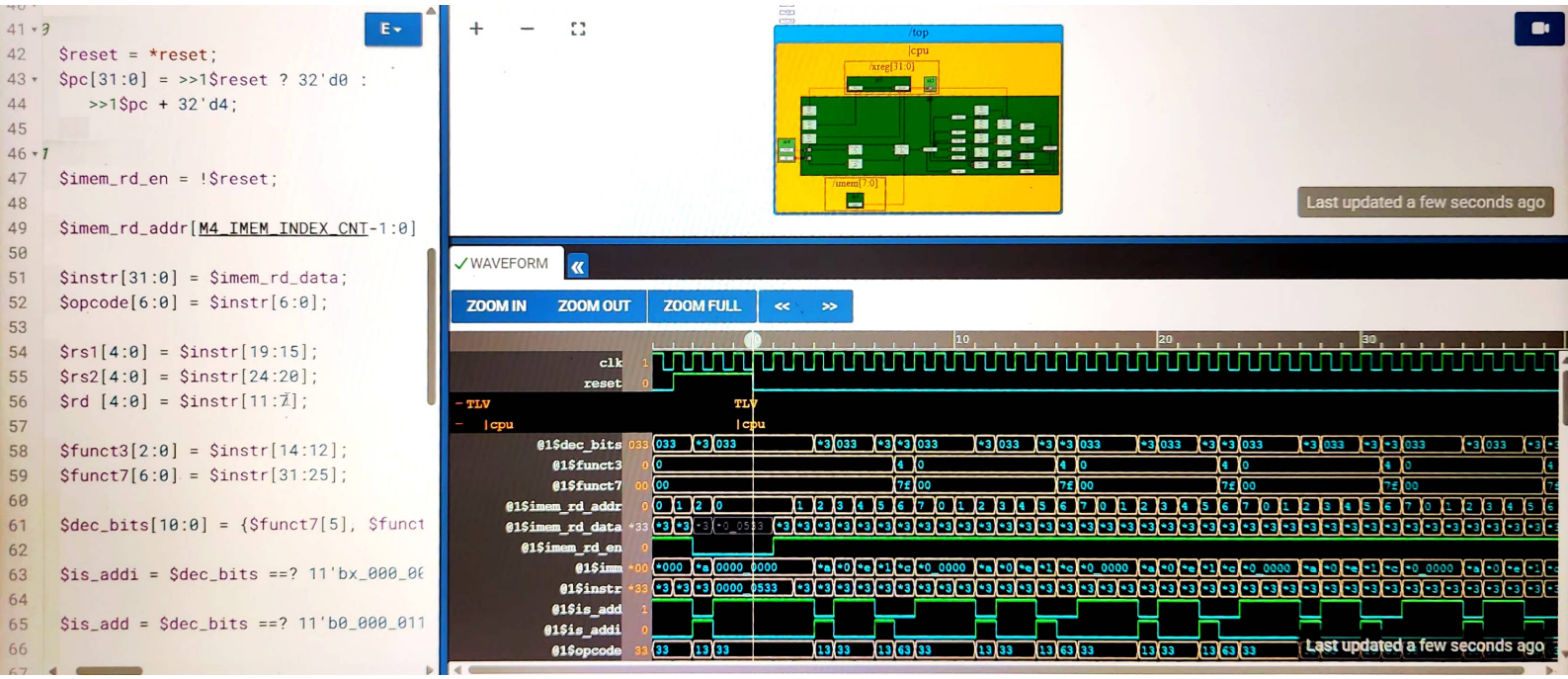

E
|cpu
@0
$reset = *reset;
$pc[31:0] = >>1$reset ? 32'd0 :
>>1$pc + 32'd4;
@1
$imem_rd_en = !$reset;
$imem_rd_addr[M4_IMEM_INDEX_CNT-1:0] =
$instr[31:0] = $imem_rd_data;
// YOUR CODE HERE
// ...
// Note: Because of the magic we are usi
// be sure to avoid having unassigi
// other than those specifically e
// Assert these to end simulation (before M
*passed = *cyc_cnt > 40;
ZOOM IN
ZOOM OUT
ZOOM FULL
<<
>>
clk
reset
TLV
|cpu
@1$imem_rd_addr
@1$imem_rd_data
@1$imem_rd_en
@1$instr
@0$pc
@0$reset
/imem[0]
|cpu/imem[0]
Last updated a few seconds ago
DIAGRAM
VIZ
/top
|cpu
@0
@1
/imem[7:0]
Last updated a few seconds ago


|cpu
@0
$reset = *reset;
$pc[31:0] = >>1$reset ? 32'd0 :
>>1$pc + 32'd4;
// YOUR CODE HERE
// ...
// Note: Because of the magic we are using
// be sure to avoid having unassigne
// other than those specifically exp
// Assert these to end simulation (before Mak
*passed = *cyc_cnt > 40;
*failed = 1'b0;
// Macro instantiations for:
// o instruction memory
// o register file
// o data memory
// o CPU visualization
|cpu
/top
|cpu
@0
@1
$pc
$reset
Last updated a few seconds ago
WAVEFORM
ZOOM IN
ZOOM OUT
ZOOM FULL
clk
reset
TLV
|cpu
@0$pc
@0$reset
SV
Last updated a few seconds ago


|cpu
@0
$reset = *reset;
$pc[31:0] = >>1$reset ? 32'd0 :
>>1$pc + 32'd4;
@1
$imem_rd_en = !$reset;
$imem_rd_addr[M4_IMEM_INDEX_CNT-1:0] = $
$instr[31:0] = $imem_rd_data;
$opcode[6:0] = $instr[6:0];
$rs1[4:0] = $instr[19:15];
$rs2[4:0] = $instr[24:20];
$rd [4:0] = $instr[11:7];
$funct3[2:0] = $instr[14:12];
$funct7[6:0] = $instr[31:25];
$dec_bits[10:0] = {$funct7[5], $funct3,
$is_addi = $dec_bits ==? 11'bx_000_00100
$is_add = $dec_bits ==? 11'b0_000_011001
/top
|cpu
/imem[7:0]
|cpu$instr
/xreg[31:0]
Last updated a minute ago
WAVEFORM
ZOOM IN
ZOOM OUT
ZOOM FULL
<<
>>
clk
reset
TLV
|cpu
@1$dec_bits
@1$funct3
@1$funct7
@1$imem_rd_addr
@1$imem_rd_data
@1$imem_rd_en
@1$imm
@1$instr
@1$is_add
@1$is_addi
@1$opcode
Last updated a minute ago


|cpu
@0
$reset = *reset;
$pc[31:0] = >>1$reset ? 32'd0 :
>>1$pc + 32'd4;
@1
$imem_rd_en = !$reset;
$imem_rd_addr[M4_IMEM_INDEX_CNT-1:0
$instr[31:0] = $imem_rd_data;
$opcode[6:0] = $instr[6:0];
$rs1[4:0] = $instr[19:15];
$rs2[4:0] = $instr[24:20];
$rd [4:0] = $instr[11:7];
$funct3[2:0] = $instr[14:12];
$funct7[6:0] = $instr[31:25];
$dec_bits[10:0] = {$funct7[5], $fun
$is_addi = $dec_bits ==? 11'bx_000_
/top
|cpu
/imem[7:0]
Last updated a few seconds ago
WAVEFORM
ZOOM IN
ZOOM OUT
ZOOM FULL
clk
reset
TLV
|cpu
@1$dec_bits
@1$funct3
@1$funct7
@1$imem_rd_addr
@1$imem_rd_data
@1$imem_rd_en
@1$instr
@1$is_addi
@1$opcode
@0$pc
Last updated a few seconds ago


$rd[4:0] = $instr[11:7];
$funct3[2:0] = $instr[14:12];
$funct7[6:0] = $instr[31:25];
$dec_bits[10:0] = {$funct7[5], $fur
$is_addi = $dec_bits ==? 11'bx_000_
$is_add = $dec_bits ==? 11'b0_000_
$rf_rd_en1 = 1'b1;
$rf_rd_en2 = 1'b1;
$rf_rd_index1[4:0] = $rs1;
$rf_rd_index2[4:0] = $rs2;
$src1_value[31:0] = $rf_rd_data1;
$src2_value[31:0] = $rf_rd_data2;
$imm[31:0] = {{21{$instr[31]}}, $ir
$imm_b[31:0] = {{19{$instr[31]}}, $
$result[31:0] = $is_add ? ($src1_
$is_addi ? ($src1_
/top
|cpu
Last updated a few seconds ago
WAVEFORM
ZOOM IN
ZOOM OUT
ZOOM FULL
clk
reset
TLV
|cpu
@1$br_cond
@1$br_tgt_pc
@1$dec_bits
@1$funct3
@1$funct7
@1$imem_rd_addr
@1$imem_rd_data
@1$imem_rd_en
@1$imm
@1$imm_b
@1$instr
Last updated a few seconds ago


$reset = *reset;
$pc[31:0] = >>1$reset ? 32'd0 :
>>1$pc + 32'd4;
$imem_rd_en = !$reset;
$imem_rd_addr[M4_IMEM_INDEX_CNT-1:0]
$instr[31:0] = $imem_rd_data;
$opcode[6:0] = $instr[6:0];
$rs1[4:0] = $instr[19:15];
$rs2[4:0] = $instr[24:20];
$rd [4:0] = $instr[11:7];
$funct3[2:0] = $instr[14:12];
$funct7[6:0] = $instr[31:25];
$dec_bits[10:0] = {$funct7[5], $funct
$is_addi = $dec_bits ==? 11'bx_000_00
$is_add = $dec_bits ==? 11'b0_000_011
/top
|cpu
/xreg[31:0]
/imem[7:0]
Last updated a few seconds ago
WAVEFORM
ZOOM IN
ZOOM OUT
ZOOM FULL
clk
reset
TLV
|cpu
@1$dec_bits
@1$funct3
@1$funct7
@1$imem_rd_addr
@1$imem_rd_data
@1$imem_rd_en
@1$imm
@1$instr
@1$is_add
@1$is_addi
@1$opcode
Last updated a few seconds ago